

ChatThread

- sender: UniversityStudent
- receiver: UniversityStudent
- message: String

```
+ ChatThread(sender: UniversityStudent,  
receiver: UniversityStudent, message: String)  
+ run(): void
```

FriendRequestThread

- sender: UniversityStudent
- receiver: UniversityStudent

```
+ FriendRequestThread(sender: UniversityStudent,  
receiver: UniversityStudent)  
+ run(): void
```

Main

```
+ main(args: String[]): void  
+ generateTestCase1(): List<UniversityStudent>  
+ generateTestCase2(): List<UniversityStudent>  
+ generateTestCase3(): List<UniversityStudent>  
+ gradeLab(students: List<UniversityStudent>, testCaseNum: int): int
```

ReferralPathFinder

- graph: StudentGraph

```
+ ReferralPathFinder(graph: StudentGraph)  
+ findReferralPath(start: UniversityStudent,  
targetCompany: String): List<UniversityStudent>
```

DataParser

```
+ parseStudents(filename: String):  
List<UniversityStudent>
```

Ballshapley

```
+ assignRoommates(students:  
List<UniversityStudent>)
```

PodFormation

```
- graph: StudentGraph  
+ PodFormation(graph:  
StudentGraph)  
+ formPods(podSize: int): void
```

Student

```
#name: String  
#age: int  
#gender: String  
#year: int  
#major: String  
#gpa: double  
#roommatePrefs:  
List<String>  
#prevInternship:  
List<String>
```

```
+ calcConnectionStrength(  
other: Student): int
```

StudentGraph

- adjList: Map<UniversityStudent, List<Edge>>

+ StudentGraph(students: List<UniversityStudent>)

+ addStudent(student: UniversityStudent) :: void

+ addEdge(student1: UniversityStudent, student2: UniversityStudent, weight: int) :: void

+ getNeighbors(student: UniversityStudent) :: List<Edge>

+ getMinNodes() :: Set<UniversityStudent>

+ displayGraph() :: void

Edge

+ neighbor: UniversityStudent

+ weight: int

+ Edge(neighbor: UniversityStudent, weight: int)

UniversityStudent

- roommate: UniversityStudent

+ UniversityStudent(name: String, age: int, gender: String, year: int, major: String, gpa: double, roommatePreference: List<String>, previousInternships: List<String>)

+ calculateConnectionStrength(other: Student) :: int

+ getRoommate() :: UniversityStudent

+ setRoommate(roommate: UniversityStudent) :: void

+ toString() :: String

Relationships:

Student ← UniversityStudent

StudentGraph ◁ ... Edge

PodFormation ◁ StudentGraph

PeterrallPathFinder ◁ StudentGraph

PeterrallPathFinder - - - -> UniversityStudent

FriendRequestThread - - - -> UniversityStudent

ChatThread - - - -> UniversityStudent

BallShapley - - - -> UniversityStudent

DataParser - - - -> UniversityStudent

Edge - - - -> UniversityStudent

StudentGraph - - - -> UniversityStudent

Main - - - -> UniversityStudent

Main - - - -> StudentGraph

Main - - - -> BallShapley

Main - - - -> PeterrallPathFinder

Main - - - -> FriendRequestThread

Main - - - -> ChatThread

Main - - - -> DataParser

Main - - - -> PodFormation

